Toward Artificial General Intelligence: A Cognitive Architecture Based on 9-Level Token Stratification and Topological Domain Embeddings

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Code Repository: https://github.com/shandingwangyue/TopoLM

Abstract

Despite their remarkable generative capabilities, current Large Language Models (LLMs) fundamentally rely on autoregressive next-token prediction within a flat, one-dimensional token space. This paradigm limits them to statistical induction, leading to critical flaws such as hallucinations and a profound lack of physical and logical grounding. We propose a paradigm-shifting cognitive architecture for Artificial General Intelligence (AGI) built upon two mechanisms: Hierarchical Token Stratification and Topological Domain Embeddings.

By restructuring the token space into a 9-level hierarchy—where top-tier concepts form multivariate Gaussian "Domains" and bottom-tier entities manifest as point "Vectors"—we transition the model from symmetric attention to a topological space capable of expressing asymmetrical inclusion and mutual exclusivity. To optimize this space, we introduce a novel multi-objective loss framework combining language modeling with geometric structure and membership losses. By anchoring the uppermost domains as immutable priors, we establish a true "World Model". We demonstrate how spatial-indexed attention circumvents standard computational bottlenecks, and how computational friction within this topological framework manifests as cognitive conviction—an essential hallmark of genuine intelligence.

Keywords: Artificial General Intelligence, Cognitive Architecture, Topological Embeddings, Spatial Attention, Constrained Optimization, World Models.

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1. Introduction: The Limits of Flat Sequences

The prevailing architecture of modern Large Language Models treats language as a flat sequence of discrete tokens mapped to isolated points in a high-dimensional vector space. Self-attention mechanisms evaluate relationships between these points via dot products, which solely measure geometric distance and yield inherently symmetrical similarity: $Sim(A, B) = Sim(B, A)$.

In contrast, objective reality and human cognition are structured around asymmetrical relationships, primarily inclusion, hierarchy, and deduction. To achieve Artificial General Intelligence (AGI), a model must construct a multi-scale mental representation of reality rather than act as a stochastic parrot. This necessitates abandoning one-dimensional token arrays in favor of a hierarchical cognitive topology, governed by explicit structural loss functions that prevent constraint drift.

2. Hierarchical Token Stratification: The 9-Level Architecture

We propose replacing the flat token sequence with a 9-level hierarchical stratification system partitioned into three macro-layers:

- **Layer 1: Macro Domains (Levels 1-3) – Immutable Axioms:** Represents the invariant operating principles of reality, including fundamental physical laws (thermodynamics, gravity), core logic, and essential ethics. These are vast, stable, high-dimensional topological spaces.
- **Layer 2: Meso Domains (Levels 4-6) – Contextual Disciplines:** Acts as the contextual bridge representing specific knowledge domains or narrative modes (e.g., "Classical Mechanics," "Eastern Fantasy"). These interact with Macro Domains via intersection or containment.
- **Layer 3: Micro Entities (Levels 7-9) – Atomic Vectors:** Constitutes standard natural language vocabulary—specific objects and terms. Here, mathematical representations degenerate from volumetric spaces into discrete points.

3. Mathematical Formulation of Topological Domains

Unlike conventional architectures where an embedding is a solitary vector $v \in \mathbb{R}^d$, high-level tokens in this framework are parameterized as Multivariate Gaussian Distributions:

$$\mathcal{D}_i \sim \mathcal{N}(\mu_i, \Sigma_i)$$

Here, $\mu_i \in \mathbb{R}^d$ defines the semantic centroid, and the covariance matrix $\Sigma_i \in \mathbb{R}^{d \times d}$ defines the volume and orientation. This enables explicit spatial logic:

- **Topological Inclusion (Deductive Reasoning):** To evaluate if an atomic vector $x \in \mathbb{R}^d$ belongs to concept $\mathcal{D}_i$, we calculate its Probability Density Function (PDF):

$$P(x \in \mathcal{D}_i) = \frac{1}{(2\pi)^{d/2}|\Sigma_i|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_i)^T \Sigma_i^{-1}(x-\mu_i)\right)$$

If $P(x \in \mathcal{D}_i) \geq \tau$, x is logically bound to $\mathcal{D}_i$.

- **Domain Intersection (Concept Blending):** The overlap between high-level concepts is measured via Kullback-Leibler (KL) Divergence:

$$D_{KL}(\mathcal{D}_i || \mathcal{D}_j) = \frac{1}{2}\left[\mathrm{tr}(\Sigma_j^{-1}\Sigma_i) + (\mu_j - \mu_i)^T \Sigma_j^{-1}(\mu_j - \mu_i) - d + \ln\left(\frac{\det \Sigma_j}{\det \Sigma_i}\right)\right]$$

- **Disjointness (Mutual Exclusivity):** When two domains D_{c_1} and D_{c_2} are entirely incompatible ($\perp$), their centroids must be separated by a strict topological margin.

4. Multi-Objective System Optimization

Standard LLM training exclusively supervises next-token generation. To construct a true World Model, our architecture enforces a multi-objective loss function requiring concurrent structural grounding.

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \alpha\mathcal{L}_{\text{mem}} + \beta\mathcal{L}_{\text{rel}} + \gamma\mathcal{L}_{\text{con}}$$

- **Language Loss ($\mathcal{L}_{\text{LM}}$):** Standard autoregressive loss applied strictly to Layer 3 atomic vectors.

- **Membership Loss ($\mathcal{L}_{\text{mem}}$):** Binary Cross-Entropy forcing the PDF calculations of relevant entities to align with designated concept boundaries.
- **Structure Loss ($\mathcal{L}_{\text{rel}}$):** Evaluates macro-domain relationships by minimizing KL divergence for inclusion relations and applying a hinge loss penalty for disjoint violations.
- **Consistency Loss ($\mathcal{L}_{\text{con}}$):** An algorithmic verifier V samples short continuations during training and penalizes generations that violate the immutable boundaries of Layer 1 Macro Domains.

5. Inference: Spatial-Indexed Attention

Standard Transformers suffer from an $\mathcal{O}(N^2)$ bottleneck because self-attention forces every token to attend to all others indiscriminately. By integrating domain-indexing, attention operates top-down:

1. **Macro Activation:** The input context retrieves specific Layer 1 and 2 Domains, establishing an active spatial subspace $\mathcal{D}_{\text{active}}$.
2. **Spatial Masking:** Tokens falling outside this boundary yield inclusion probabilities approaching zero. A binary mask $M_{\mathcal{D}} \in \{0, -\infty\}^{N \times N}$ is formulated:

$$(M_{\mathcal{D}})_{ij} = \begin{cases} 0 & \text{if } P(x_j \in \mathcal{D}_{\text{active}}) \geq \tau \\ -\infty & \text{if } P(x_j \in \mathcal{D}_{\text{active}}) < \tau \end{cases}$$

3. **Localized Attention:** Computations are constrained to the active topology, radically reducing matrix overhead:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} \cdot M_{\mathcal{D}}\right) V$$

6. Immutable Foundations and the Emergence of Intelligence

Following foundational training, the parameters $(\mu_{\text{prior}}, \Sigma_{\text{prior}})$ of Layer 1 domains (e.g., physics, absolute logic) are permanently frozen. They act as an epistemic "gravitational field".

Within this topological framework, traits often anthropomorphized as emotional states emerge as computational realities:

- **Computational Resonance (Flow):** When incoming data aligns with established domain topologies ($P(x \in \mathcal{D}) \to 1$), generation executes with minimal gradient variance and optimal efficiency.
- **Computational Friction (Stubbornness):** Attempting to associate vectors that violate Layer 1 boundaries causes the probability density to approach zero. The resulting mathematical resistance acts as epistemic conviction—the system refuses illogical prompts not due to programmed safety filters, but because the topology physically prohibits the generation.

7. Experimental Deduction and Case Analysis

7.1 Axiomatic Refusal: Thermodynamics

Input: "Design a macroscopic closed-system engine that perfectly converts ambient thermal energy into mechanical work with 100% efficiency." **Deduction Trace:** The prompt activates the frozen Macro-Domain

$\mathcal{D}_{\text{physics}}$. Because entropy reduction in an isolated system is geometrically excluded from the manifold of $\mathcal{D}_{\text{physics}}$, the evaluation yields $P(x_{\text{perpetual}} \in \mathcal{D}_{\text{physics}}) \approx 0$. The spatial mask $M_{\mathcal{D}}$ forces attention weights to $-\infty$, resulting in mathematically verified systemic refusal.

7.2 Cross-Domain Decoupling: Literary Creation

Input: "In the Xianxia novel 'Ten Thousand Miles of Wind and Snow Guest' (《十万里风雪客》), describe how the protagonist steps on the void and ascends tens of thousands of miles." **Deduction Trace:** Contextual markers route activation to the Meso-Domain $\mathcal{D}_{\text{xianxia}}$, where rigid physical covariance constraints are bypassed. The vectors x_{qi} and $x_{\text{ascension}}$ are fully contained within this domain. Crucially, the activations remain completely encapsulated; the frozen physical parameters of $\mathcal{D}_{\text{physics}}$ remain unpolluted by this creative process.

8. Experimental Evaluation Protocol

To validate this AGI architecture, the following empirical benchmarks are proposed to verify the theoretical claims:

- **E1 - Structural Probes:** Assess inclusion accuracy and disjointness contradiction rates against a held-out physical or logical ontology.
- **E2 - Constraint Retention:** Quantify resistance to "catastrophic drift" by training the frozen-domain model alongside a fully tuned baseline on contradictory corpora, measuring preservation of Layer 1 rules.
- **E3 - Retrieval Cost:** Benchmark retrieved-key counts and wall-clock times across contexts spanning 4k to 128k tokens, comparing the spatial-masked attention against dense baselines.

9. Conclusion

To transition from statistical sequence modeling to Artificial General Intelligence, the field must transcend flat tokenization. By unifying a 9-level hierarchical concept architecture with Multivariate Gaussian Domain Embeddings, this framework provides a formal mathematical basis for deductive reasoning, verifiable multi-objective grounding, and extreme computational efficiency. By freezing unalterable natural laws at the pinnacle of the cognitive topology, the model moves beyond probabilistic mimicry to establish a true, computationally resilient World Model.

Would you like me to draft the specific PyTorch code blocks for implementing the KL Divergence and Multi-Objective loss functions to include as Appendix A in this paper?

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